CHAPTER 15

ION BEAM MACHINING (I B M)

15.1 INTRODUCTION

IBM is an electro-thermal process. This process removes the materials unlike the EDM & EBM where bombardment of electrons occur like bombardment of the hot ions as in PAM or bombardment of photons in LBM, which result in intense heating, melting and vapourisation of work material.

15.1.1. Definition

IBM is a material removal process in which the material is removed by the sputtering of ions ie., the surface atoms are dislodged from the incident bombarding ions by the Transfer of Kinetic energy.

15.2. EQUIPMENT

The equipment consists of an electron gun from which free electrons are discharged into a vacuum chamber filled with argon gas which is then ionized by the electrons. Thus a stream of ions is generated which is directed onto the work piece surface. The electron gun consists of an electrode, the geometry of which may be spherical or cylindrical or cylindrically curved to obtain or focus the beam of electrons as a spot or line or a plane or convexing downwards to produce a parallel beam. A best beam shape is obtained by adjusting the anode guard ring.

The gun is made up of stainless steel because of its low thermal conductivity. The electrons released from the cathode are accelerated towards the gas chamber by the anode by supplying a suitable anode potential.

High vacuum of the order of 10^{-4} Torr is required to ionize the gas inside the chamber. High vacuum pumps are used for this purpose.

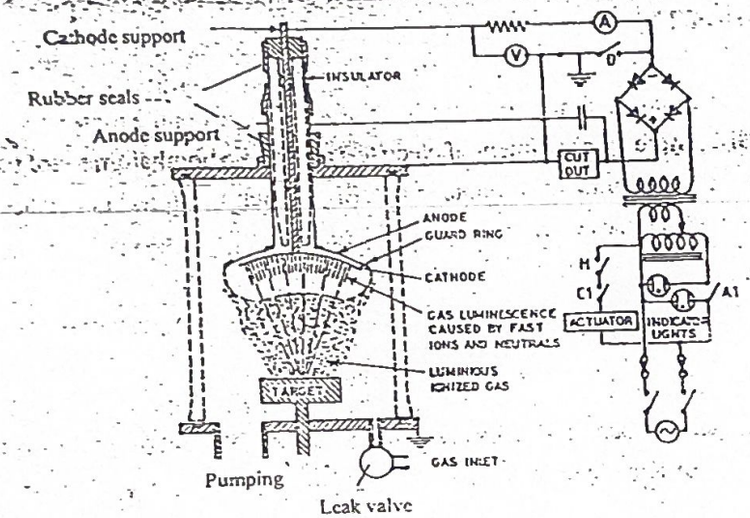

Fig. 15.1. Schematic diagram of Ion Beam Maching

The geometry of the electrode itself will focus the ion beam on to the work piece surface with a high velocity to dislodge the surface atoms.

15.2. WORKING PRINCIPLE

When voltage is applied across the electrodes in the order of 1KV to 100 KV range, kept in an vacuum chamber filled with a gas like argon, flow-discharge occurs, ie,, the gas atoms are knocked to strip off their electrons, which result in the formation of ions, by the electron beam produced in the cathode. The ionized gas emits positive ions which are accelerated towards the cathode. The fast moving ions collide with the gas molecules and the ions gets excited and go to the next higher energy state releasing secondary electrons. The trajectories of these ions and electrons are normal to the cathode surface, and because of the cathode curvature, they are focussed on to a particular

spot. When the stream of ions bombard the work piece material, they dislodge the surface atoms of the work material by transferring their entire kinetic energy. The number of atoms dislodged by each bombarding ion is called the sputtering coefficient or sputtering yield (Y)

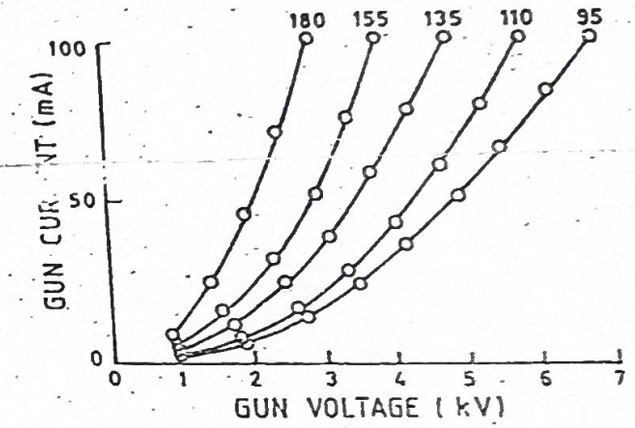

Figure 15.2. Current and voltage dependence of pressure

$$Y = \frac{\text{Number of atoms removed}}{\text{Number of striking ions}}$$

This coefficient can increase if the atomic weight of the bombarding ions is more. Moreover it is also influenced by the angle of incidence of the ion and the material being bombarded. Little heat is only generated in ion beam machining.

15.4. ADVANTAGES AND DISADVANTAGES

15.4.1. Advantages

1. The equipment and the process are very simple.
2. No chemical reagents and etchants are required.

3. Thin films can be easily processed.
4. The etching rates are controllable
5. The process is almost universal.
6. There is no undercutting as in the case of chemical etching.

15.3.2. Disadvantages

1. The process is relatively expensive.
2. Slow material removal rate.
3. Possibility for thermal and radiation damages.
4. Needs extra accessories like vacuum pump and gas circulating pumps.

15.5. APPLICATIONS

1. Used for the micro-machining of electronic components like Integrated circuits, figuring optical surfaces and fabrication of fine wire dies in refractory materials.
2. Shallow contouring and deep cutting are possible.
3. Glass, alumina, quartz, silica, agates, porcelain cermets, mixtures of quartz and asbestos, metals and oxides can be machined or can be processed without any difficulty.
4. Cleaning of surface contaminations, polishing of glass and sputter deposition ie, thin film metal deposition are possible
5. Etching of specimens before studying the micro structure calls for IBM application.
6. Mainly used by the electronic industries for multi-layer fabrication of Integrated circuits.